

Mohammed Abdul Rahman

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EDUCATION

Texas A&M University

M.S. in Computer Science and Engineering (CGPA 3.5/4.0)

Corpus Christi, TX

(May 2025)

- Coursework: Design & Analysis of Algorithms, Machine Learning, Deep Learning and Advance Information Assurance.

PROJECTS

Dine Easy (MERN Stack)

- **Tech Stack:** React.js, Node.js, Express.js, MongoDB
- Developed a responsive, user-friendly interface with React.js, improving user interaction by 30% through real-time menu updates and seamless navigation.
- Built a RESTful API using Node.js and Express.js to ensure efficient data communication; integrated JWT-based authentication, boosting login security by 40%.
- Implemented order management features with MongoDB, scaling performance to handle 10,000+ users while maintaining secure data storage.

Predicting Energy Usage of Electrical Appliances using ML

- **Tech Stack:** Python, Scikit-learn, Pandas, NumPy
- Engineered advanced predictive models for residential energy consumption, leading to a 25% increase in forecast accuracy; provided actionable insights that optimized energy-saving initiatives for homeowners and reduced overall consumption by 15%.
- Performed comprehensive regression model comparison to identify the optimal model for energy prediction efficiency.

Quick Chat (MERN Stack)

- **Tech Stack:** MongoDB, Express.js, React.js, Node.js, Socket.IO
- Built a real-time chat application using the MERN stack, enabling seamless and instant communication access time by 50% between users.
- Implemented WebSocket communication via Socket.IO for real-time messaging & updates, for a smooth UX.
- Designed a responsive and user-friendly interface using React, with a focus on intuitive navigation & chat accessibility.
- Integrated MongoDB to streamline message storage and retrieval processes, resulting in a dramatic reduction of chat history access time by 50%, enhancing user engagement during real-time conversations.
- Secured the application with JWT-based authentication, ensuring strong user data protection.
- Optimized backend communication through Express.js and Node.js to ensure robust and scalable server-side logic.

SKILLS

- **Programming Languages:** Java, Python, C, JavaScript, SQL
- **Web Development:** React.js, Node.js, Express.js, MongoDB, HTML5, CSS
- **Frameworks/Libraries:** MERN Stack, TensorFlow, Scikit-learn, NumPy, Pandas
- **Tools:** Git, GitHub, PowerBI, MS Office, Tableau
- **Soft Skills:** Leadership, Problem Solving, Team Collaboration, Agile Methodology

CERTIFICATIONS

- Java Programming Masterclass for Software Developers - Udemy
- Machine Learning with Big Data by the University of California San Diego

ACHIEVEMENTS

- Organized and participated in university coding competitions and hackathons.
- Directed multiple social service activities, leading teams of 10+ in delivering impactful community services